MAT1100 Homework 3

Aaron Avram

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Question 2

(a) *Proof.* Let $P \curvearrowright N$ via conjugation, since $N \neq \{e\}$ it is a p-group and thus the set of fixed points $N^P = 0 \mod p$ and since $e \in N^P$ this implies that $N^P \neq \{e\}$. Now

$$N^P = \{x \in N : \forall p \in P \text{ s.t. } pxp^{-1} = x\} = Z(P) \cap N$$

hence $Z(P) \cap N \neq \{e\}$. □

(b) *Proof.* We proceed by induction. Namely we prove the following statement by induction on $j \in \mathbb{N} \cup \{0\}$: If P is a p-group such that $N \leq P$ and $p^j \mid |N|$ then N has a normal subgroup of order p^{j-1} that is also normal in P.

The base case is trivial as $p^0 = 1$ and the trivial subgroup $\{e\}$ has the required properties.

Now for the inductive step, suppose that the statement holds for some $j \in \mathbb{N} \cup \{0\}$. Now consider a p-group P and a normal subgroup N such that $p^{j+1} \mid |N|$. Since $j \geq 0$ this implies that $N \neq \{e\}$ and by (a) this implies that $Z(P) \cap N$ is a non-trivial p-group. Thus by Cauchy's theorem there is $x \in Z(P) \cap N$ of order p. Now $\langle x \rangle \lhd N, P$. Moreover $P/\langle x \rangle$ is a p-group with normal subgroup $N/\langle x \rangle$ and $p^j \mid |N/\langle x \rangle|$. Thus by the inductive hypothesis $N/\langle x \rangle$ has a normal subgroup $\overline{H}$ of order p^j that is also normal in $P/\langle x \rangle$. By the correspondence isomorphism theorem, if π is the natural projection for $P \to P/\langle x \rangle$ then $H = \pi^{-1}(\overline{H})$ is a normal subgroup of N of order p^{j+1} which is also normal in P as required. □

(c) *Proof.* Suppose G is a group, then for any prime p applying (b) to a Sylow p-subgroup of G shows that for any $p^j \mid |G|$, G has a subgroup of order p^j. However, such a subgroup need not be normal in G. For example take S_3 and the 2 element subgroup $\{e, (12)\}$. This is not normal as

$$(23)(12)(23)^{-1} = (13) \notin \{e, (12)\}$$

□

(d) *Proof.* Given a p-group P with $|P| = p^a$, we prove the following stronger statement by induction on $i \in \mathbb{N}$:

$$|Z_i(P)| > p^{i-1}$$

and when $i = a$ this implies that $|Z_a(P)| > p^{a-1}$ and hence $|Z_a(P)| = p^a$ by Lagrange's theorem which is equivalent to $Z_a(P) = P$ as required.

For the base case $i = 1$: $|Z_1(P)| = |Z(P)| > p^0 = 1$ as a p-group has a non-trivial center.

Now suppose the statement holds for some $i \in \{0, \ldots, a-1\}$. Then

$$Z_{i+1}(P) = \pi_i^{-1}(Z(P/Z_i(P)))$$

hence

$$|Z_{i+1}(P)| = |Z_i(P)||Z(P/Z_i(P))|$$

now first suppose that $Z(P/Z_i(P)) = \{e\}$, this implies that $P/Z_i(P) = \{e\}$ as non-trivial p-groups have non-trivial centers. Thus $P = Z_i(P)$ and then clearly $Z_{i+1}(P) = P$ which clearly suffices.

Otherwise, $Z(P/Z_i(P)) \neq \{e\}$ and hence $|Z(P/Z_i(P))| \geq p$ and thus

$$|Z_{i+1}(P)| = |Z_i(P)||Z(P/Z_i(P))| > p^i p = p^{i+1}$$

as required. □

(e) *Proof.* To prove that a nilpotent group G is solvable it suffices to show that

$$Z_{i+1}(G)/Z_i(G)$$

is abelian then reversing the upper central series is clearly a solvable series for G. Now

$$Z_{i+1}(G) = \pi_i^{-1}(Z(G/Z_i(G)))$$

so by definition

$$Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G))$$

hence $Z_{i+1}(G)/Z_i(G)$ is abelian and G is solvable.

For an example of a group that is solvable but not nilpotent consider S_3.

$A_3 \cong C_3$, $S_3 \rhd A_3 \rhd \{e\}$ is a solvable series for S_3. However $Z(S_3) = \{e\}$ and thus $Z_1(S_3) = Z(S_3) = \{e\}$ meaning that $Z_i(S_3) = \{e\}$ for all $i \in \mathbb{N}$. Hence S_3 is not nilpotent. □